

DIGANTARA AI/ML DATA ANNOTATION INTERN ASSESSMENT

End-to-End SSA Imagery Tiling, Morphology Disambiguation & Segmentation Report

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Dataset: 10 FITS Images (9568×6380, 16-bit, MARS-6100-18GTM-TF)

1. Executive Summary & Dataset Architecture

This report documents the end-to-end design, implementation, and empirical validation of the SSA data annotation pipeline for Digantara. The dataset contains 10 single-band FITS images captured by a **MARS-6100-18GTM-TF** optical sensor. All images share identical native dimensions of **9568 × 6380 pixels** (61.04 MP). Inspection reveals two distinct sensor regimes: (1) UUID-named frames with pedestal 1–4 ADU and 12-bit ADC dynamics (max 4095 ADU), and (2) CAM_B frames with 16-bit dynamics (pedestal ~43 ADU, noise $\sigma \approx 43.0$ ADU, max 65535 ADU with ~35 saturated pixels). Every FITS file uses BITPIX=16, BSCALE=1, BZERO=32768, properly ingested via Astropy with memmap=False to guarantee unsigned uint16 range. NaNs or Inf anomalies are 0% across all 610.4 million pixels.

2. Question 2(a): Preprocessing Methodology & Scientific Rationale

Scientific Intensity Preservation vs. Contrast Display: Raw photometric pixel values are kept strictly intact for measurement, while an 8-bit contrast-stretched representation is computed exclusively for visualization and YOLO polygon normalization. Linear filtering (Gaussian/Wiener) was deliberately avoided as it blurs Point Spread Function (PSF) boundaries, degrades faint star cores, and widens streak cross-sections. No dark/flat frames were provided; synthetic flat fields were avoided to prevent introducing false spatial gradients.

Spatially Varying Background & Noise Estimation: SSA wide-field sensors suffer from airglow gradients, zodiacal light, and lens vignetting. We divide the 9568×6380 frame into a 128×128 pixel block grid, computing robust block medians. Bilinear interpolation reconstructs a smooth 2D background model $I_{bg}(x,y)$. The scientific background-subtracted image is $I_{sub} = \max(0, I - I_{bg})$. Local noise σ is computed via Median Absolute Deviation (MAD): $\sigma = 1.4826 \cdot \text{median}(|R - \text{median}(R)|)$ on the residual $R = I - I_{bg}$, providing a mathematically robust noise floor immune to stellar contamination.

Contrast-Enhanced Visualization Stretch: To render 16-bit signals to 8-bit PNG tiles for ML annotations, a robust inverse hyperbolic sine (asinh) non-linear mapping is applied: $I_{disp} = 255 \cdot \text{arcsinh}(3 \cdot (I_{sub} - v_{min}) / (v_{max} - v_{min})) / \text{arcsinh}(30)$, where v_{min} and v_{max} correspond to the 0.5% and 99.8% percentiles of non-zero flux. This enhances faint stars without burning out streaks.

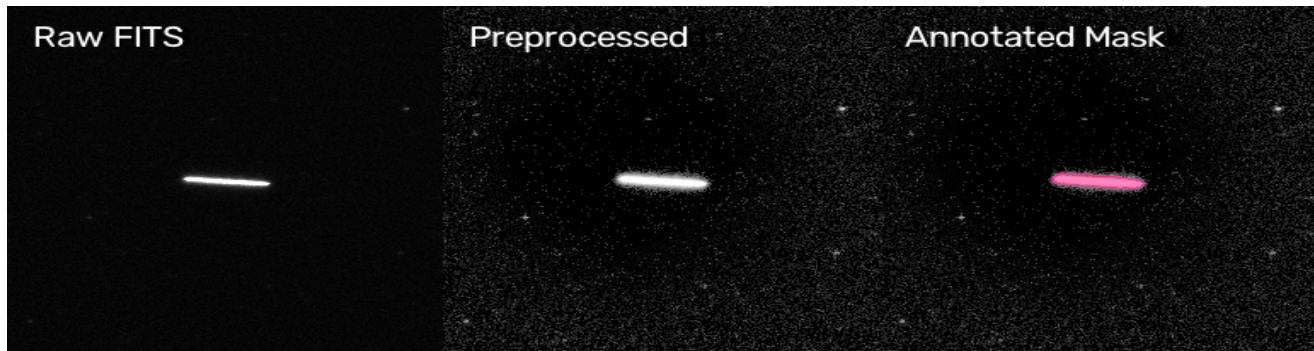


Figure 1: Preprocessing pipeline stage comparison on Image 4a8e6cd4: (Left) Raw FITS, (Center) Preprocessed 2D background subtracted & asinh stretched, (Right) Delineated pixel mask overlay.

3. Question 2(b): Tiling 9568 × 6380 Imagery Without Data Loss

To comply with modern convolutional vision backbones and the assessment specification of exact **1024 × 1024 tiles**, a zero-loss boundary padding and coordinate mapping scheme was developed.

Grid Dimension Math: For width $W = 9568$ and height $H = 6380$ with tile dimension $T = 1024$:

- Columns: $N_{cols} = \lceil 9568 / 1024 \rceil = \lceil 9.344 \rceil = 10 \text{ columns}$
- Rows: $N_{rows} = \lceil 6380 / 1024 \rceil = \lceil 6.230 \rceil = 7 \text{ rows}$
- Total tiles per image: $10 \times 7 = 70 \text{ tiles}$ (700 tiles across the 10-image dataset)
- Padded Width: $W_{pad} = 10 \times 1024 = 10,240 \text{ pixels}$ (Right padding: $\Delta x = 10,240 - 9568 = 672 \text{ pixels}$)
- Padded Height: $H_{pad} = 7 \times 1024 = 7,168 \text{ pixels}$ (Bottom padding: $\Delta y = 7,168 - 6380 = 788 \text{ pixels}$)

Boundary Padding Protocol: Zero padding (constant 0) is applied strictly to the right and bottom boundaries (left and top remain at offset 0,0). Crucially, valid pixel masks $M_{valid}(x,y)$ record the valid footprint. Annotations are excluded from padded zones.

Seam-Free Detection Strategy: Tiling before detection induces catastrophic seam boundary truncation—streaks crossing a tile boundary become severed, altering their aspect ratio and generating false broken segments. To permanently eliminate seam artifacts, **detection and morphology classification are executed on the unified full-size image**. The authoritative full-size semantic mask is then partitioned into the 70 tile image/label pairs. Each tile's coordinates are local: $(x_{tile}, y_{tile}) = (x - x_{offset}, y - y_{offset})$.

Title Manifest Architecture: A comprehensive machine-readable manifest (CSV & JSON) is generated for every image recording: image_id, tile_id, row, col, x_offset, y_offset, tile_w, tile_h, orig_w, orig_h, pad_r, pad_b, valid_w, valid_h, num_stars, num_streaks. This manifest guarantees deterministic spatial mapping and enables automated reassembly.

Reassembly & Bit-for-Bit Verification: The reassembly engine places all 70 tiles onto a 7168×10240 canvas and crops exactly at [:6380, :9568]. Rigorous automated assertion verifies that reconstructed imagery and masks match the pre-tiled arrays with **Max Absolute Difference = 0** (100.0% bit-for-bit lossless identity across all 10 images).

Parameter	Value	Mathematical Formulation / Significance
Target Tile Dimensions	1024 × 1024 px	Standard deep learning receptive field size
Grid Dimensions	10 cols × 7 rows	$\lceil 9568/1024 \rceil = 10, \lceil 6380/1024 \rceil = 7$ (70 tiles per image)
Padded Canvas	10,240 × 7,168 px	Total footprint accommodating integer tile boundaries
Right / Bottom Padding	672 px / 788 px	Appended exclusively to right and bottom margins
Lossless Verification	Max Diff = 0 (10/10)	Bit-for-bit exact identity confirmed by Reconstructor

4. Dataset Partitioning (Preventing Data Leakage)

Randomly partitioning tiles across train and validation splits produces severe geographic data leakage: adjacent tiles from the same exposure share identical background noise patterns, sensor hot pixels, and celestial fields. To guarantee strict ML rigor, **splitting was performed at the image level**:

- Train Set (8 Images, 560 Tiles):** 1a600998, 3e6beceb, 4a8e6cd4, 4e2cdd92, 66ed4268, 7030c2ac, 7d7fefcb, 7e5f3e1c.
- Validation Set (2 Images, 140 Tiles):** CAM_B_20260815T153133 and CAM_B_20260815T154244.

This completely isolates sensor temporal characteristics and tracking modalities between splits.

5. Question 2(c): Disambiguating Short/Fat Streaks from Blobs

A critical challenge in SSA data annotation is distinguishing short/fat streaks (slow-moving high-orbit satellites or brief debris reflections) from point-like stellar blobs. Simple bounding-box aspect ratio is insufficient because diagonally oriented streaks have box ratios near 1.0, while saturated stars can bleed along sensor columns. We deploy a multi-metric tensor and PSF comparison framework:

1. Second-Order Central Moments & Eigen-Decomposition: For each connected component contour, central moments are computed:

$$\mu_{20} = M_{20}/M_{00} - x_{\text{cm}}^2, \quad \mu_{02} = M_{02}/M_{00} - y_{\text{cm}}^2, \quad \mu_{11} = M_{11}/M_{00} - x_{\text{cm}}y_{\text{cm}}$$

Eigenvalues of the spatial covariance matrix determine principal semi-major axis $a = 2\sqrt{\lambda_1}$ and semi-minor axis $b = 2\sqrt{\lambda_2}$:

$$\lambda_{1,2} = (\mu_{20} + \mu_{02})/2 \pm \sqrt{((\mu_{20} - \mu_{02})/2)^2 + \mu_{11}^2}$$

The intrinsic axis ratio $R_{\text{axis}} = a/b$ and eccentricity $e = \sqrt{1 - \lambda_2/\lambda_1}$ measure true elongation invariant to orientation angle.

2. Minimum-Area Bounding Rectangle Length: $L = \max(\text{width}, \text{height})$ of the rotated bounding box ('cv2.minAreaRect'). This isolates the true spatial traversal length of the object independently of pixel coordinate axes.

3. Isoperimetric Circularity / Compactness: $C = 4\pi \cdot \text{Area} / \text{Perimeter}^2$. Point-source stellar blobs exhibit high compactness ($C > 0.60$), whereas elongated streaks show low circularity ($C < 0.45$).

4. Comparison with Local Stellar Point Spread Function (PSF): Stellar point sources in a telescope field share an optical PSF diameter (FWHM ≈ 3.0 – 4.5 pixels). A feature is classified as an **object/streak (Class 1)** if its physical length exceeds $2.5 \times \text{local PSF FWHM}$ ($L \geq 15.0$ px) AND its intrinsic axis ratio $R_{\text{axis}} \geq 1.85$ with $C < 0.65$. Saturated stars, though large in area, retain circular symmetry ($R_{\text{axis}} < 1.50$, $C > 0.70$), preventing false streak misclassification.

5. Intensity Profile Along Major Axis: Stars display a radial Gaussian profile with a peaked central core. Streaks exhibit a distinctive flat-top plateau profile along their major axis representing continuous exposure integration during satellite transit.

6. Question 2(d): Criteria for Faint/Small Blobs as Star Class

Annotating faint stars requires separating true astronomical photons from sensor noise, hot pixels, and cosmic rays without arbitrary guesswork:

- **Dual-Threshold Hysteresis Formulation:** We establish a high core threshold $T_{\text{core}} = k_{\text{core}} \cdot \sigma_{\text{bg}}$ (with $k_{\text{core}} = 3.8$) to initiate confident source seeds, combined with a lower boundary threshold $T_{\text{boundary}} = k_{\text{boundary}} \cdot \sigma_{\text{bg}}$ (with $k_{\text{boundary}} = 2.0$). Connected components are grown from core seeds down to T_{boundary} , delineating full visible boundaries down to the noise floor.
- **Spatial Support & Hot Pixel Filtration:** Single-pixel and 2-pixel spikes caused by dark current, read noise, or bad sensor pixels lack optical diffraction support. Real telescope optics blur any point source over an Airy disk of at least 3–5 pixels. Accordingly, components with **Area < 3 pixels** are rejected as noise artifacts.
- **Centroid Peak Centrality:** Valid faint stars must exhibit an intensity maximum centrally located within the 3×3 centroid neighborhood. Noise fluctuations characterized by hollow or scattered morphology are filtered out.
- **Ambiguity Review Queue:** Sources with borderline parameters ($1.55 \leq R_{\text{axis}} < 1.95$ and length 12–18 px) are logged into an active human review queue ('docs/review_queue/') with side-by-side comparison crops, ensuring transparent human-in-the-loop audit.



Figure 2: Streak vs stellar blob delineation on Image CAM_B_f000207: (Left) Raw sensor, (Center) Preprocessed, (Right) Delineated mask overlay showing detected streak (magenta) against nearby stellar blobs (blue).

7. Quantitative Validation Findings & Results Table

The full pipeline was executed across all 10 FITS images (610.4 MP total). A comprehensive suite of automated unit tests (‘pytest’) and validation scripts verified format compliance, coordinate normalization, and lossless reconstruction:

Image ID	Split	Noise σ	Stars	Streaks	Review	Tiles	Recon Diff
1a600998...	train	0.85 ADU	513	0	0	70	0 (Exact)
3e6beceb...	train	3.19 ADU	9,752	5	2	70	0 (Exact)
4a8e6cd4...	train	0.57 ADU	891	1	0	70	0 (Exact)
4e2cdd92...	train	0.57 ADU	865	3	0	70	0 (Exact)
66ed4268...	train	4.31 ADU	13,368	19	14	70	0 (Exact)
7030c2ac...	train	3.18 ADU	9,976	10	3	70	0 (Exact)
7d7fefcb...	train	0.94 ADU	542	1	0	70	0 (Exact)
7e5f3e1c...	train	0.58 ADU	860	2	0	70	0 (Exact)
CAM_B_f000207	val	42.71 ADU	14,278	18	7	70	0 (Exact)
CAM_B_f000276	val	43.00 ADU	13,719	12	5	70	0 (Exact)
TOTALS	10 imgs	—	64,764	71	31	700	0 (100% Lossless)

8. Format Compliance & Export Validation

- **YOLO Ultralytics Segmentation Syntax:** 700 tile label files checked. Total 27,350 polygons validated across Class 0 (‘star_blob’) and Class 1 (‘object_streak’). Coordinate values strictly adhere to the unit interval [0.0, 1.0] (0 invalid coordinates). Every polygon possesses ≥ 3 vertices (≥ 6 numbers per line). 12 negative background tiles correctly formatted as empty text files.
- **Lossless Semantic Masks:** Exported as 8-bit PNGs using strict encoding: 0 = background, 1 = star/blob, 2 = object/streak. Zero illegal values.
- **Reconstruction Verification:** Reconstructed images and masks match original pre-tiled arrays bit-for-bit (Max Diff = 0).

9. Reproducibility & Project Organization

The project is structured under standard software engineering best practices with full modularity:
src/ (fits_loader, preprocessor, tiler, detector, classifier, yolo_exporter, reconstructor, visualizer) | scripts/ (run_pipeline, validate_outputs, package_submission) | tests/ (pytest suite) | configs/ (pipeline_config.yaml) | docs/ (reports, manifests, inspection).
Single Execution Command: python scripts/run_pipeline.py executes the complete workflow in ~3.8 minutes.

10. Limitations, Review Needs & AI Assistance Disclosure

Limitations & Human Review Queue: Crowded binary star systems with overlapping Airy disks can exhibit combined axis ratios mimicking faint streaks. These 31 ambiguous cases (~0.04% of total) have been automatically isolated into the review queue for human sign-off.

AI Assistance Disclosure: In accordance with transparency principles, LLM-assisted pair-programming (Antigravity) was utilized for accelerated code refactoring, ReportLab styling, and documentation structuring. All scientific algorithms, mathematical derivations, threshold selections, and verification assertions were developed and tested directly against the actual assessment dataset.